

Elisa VALENTE

R&D Tech Lead · Data Scientist

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Viana do Castelo, Portugal

mariaevalente.github.io mariaevalente in elisavalente

Portuguese (native) and English (C1)



Data scientist and R&D lead with over five years in building energy. I take machine-learning ideas from the first experiment to things people rely on: control rules in the production HVAC controller, the savings baselines used for invoicing and automated monthly reports for clients. First inventor of an international patent application. Seven years of competitive athletics taught me the discipline I bring to every project.

WORK EXPERIENCE

Ongoing
May 2026

R&D Tech Lead, BANDORA, Portugal

- > Lead the R&D strategy, roadmap and data science team, and mentor teammates, interns and master's students.
- > Working with the heads of every department, identified company-wide gaps in data visibility and built a data observability and quality platform, with automated monthly reports that show each client how our controllers perform (time-series reads cut from 13 s to 3 ms).
- > Technical lead of an R&D funding application (2026).

Leadership Roadmap Mentoring Data Quality Observability Streamlit Docker SQL MongoDB Claude Code

May 2026
March 2021

Data Scientist, BANDORA, Portugal

- > Owned the full data science cycle for HVAC optimisation: problem definition, synthetic data from EnergyPlus digital twins, modelling (LSTM, XGBoost), experiments tracked with MLflow and DVC, and evaluation.
- > First inventor of international patent application [WO 2026/033501](#): pre-training forecasting models on digital-twin data so that control works from day one. Presented at Ai4 (Las Vegas, 2024) and AI Nest (Coimbra, 2026).
- > Turned findings from the HVAC optimisation research into energy and comfort rules for the production rule-based controller, which kept 13 client sites 12% below their ASHRAE Guideline 14 baselines.
- > Ran PoCs to de-risk new approaches. The latest, a predictive controller tested on 10 simulated buildings, used 26% less HVAC energy than fixed schedules and cut discomfort by 46% against rule-based control.
- > Designed and built BuildFlow, a domain-specific language for expert HVAC control rules (lexer, parser, compiler, interpreter), and a simulation platform where one TOML file describes any building.
- > Designed the savings baseline used for invoicing and supervised its move to an ASHRAE Guideline 14 model tournament (31 of 32 buildings compliant). Supervised an NLP project (TF-IDF) that classifies raw device data points with 86% accuracy.

Python pandas scikit-learn Keras XGBoost LSTM NLP Time Series Forecasting Optimisation EnergyPlus
MLflow DVC Azure AWS S3

August 2020
February 2020

Researcher, UNIVERSITY COLLEGE LONDON, England

Network Anomaly Detection using Adversarial Deep Learning

- > Development of algorithms to allow the detection of network attacks using adversarial deep learning methods. (*Master Dissertation*)

Python Anomaly Detection Generative Adversarial Networks Artificial Neural Networks Tree-Based Models

EDUCATION

December 2020
September 2015

BSc. and MSc. Computer Science and Engineering, UNIVERSITY OF MINHO, Braga, Portugal

- > Specialization in Data Science and Intelligent Systems
- > Also a secondary specialization in NoSQL Databases and Computer Systems Security

Final Grade: 16/20

CERTIFICATES

October 2024 | Design Patterns in Python, UDEMY, Online
(Certificate)

September 2022 | Machine Learning Engineering for Production (MLOps), COURSERA, Online
April 2022 | Four courses: ML in Production, Data Lifecycle, Modeling Pipelines, Deployment (certificates 1, 2, 3, 4)

SKILLS

Languages	Python, SQL; also R, Java, JavaScript
Machine learning	pandas, NumPy, scikit-learn, Keras, XGBoost, time series, forecasting, optimisation, NLP
Simulation	EnergyPlus, digital twins
Databases	PostgreSQL, MongoDB, MySQL
Engineering	Git, Docker, MLflow, DVC, Streamlit, Plotly, Claude Code
Cloud	Azure, AWS S3
Tools	Jira, Bitbucket, GitHub, Power BI, LaTeX